

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (CSE) (Credit System) Degree Examinations

May - June 2023

20CS603 - COMPUTER GRAPHICS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Draw the necessary **diagram** wherever required.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Briefly explain any 3 applications of Computer Graphics.	10	L*2	1	1
	b) With a neat diagram, explain the architecture of raster scan display system.	10	L2	1	1
2.	a) Explain Sutherland Hodgeman polygon clipping algorithm with a relevant example.	10	L2	1	1
	b) Explain the Cohen Sutherland line clipping algorithm in detail.	10	L2	2	1
3.	a) What is concatenation transformation? Explain rotation about a fixed point.	8	L2	2	1
	b) Define and represent the following 2-D transformations in homogenous coordinate system of Translation, Rotation and Scaling.	12	L2	2	1
Unit – II					
4.	a) Explain the basic transformations in 3D and represent them in matrix form.	12	L2	3	1
	b) Discuss following color cube models in detail.				
	i) RGB Color model	8	L2	4	1
	ii) CMY color model				
5.	a) Explain following shading models:				
	(i) Constant Shading				
	(ii) Interpolated shading				
	(iii) Polygon mesh shading	10	L2	4	1
	b) Explain the Phong illumination model. Indicate the advantages and disadvantages.	10	L2	3	1
6.	a) Derive mathematical formulation for uniform B-spline and explain the blending function.	10	L5	3	1
	b) Discuss different types of projection in detail.	10	L2	3	1
Unit – III					
7.	a) Explain the Scanline Polygon filling algorithm.	10	L2	4	1
	b) How depth sort method is used to identify visible surfaces. With neat sketches discuss the different cases of depth overlap.	10	L2	4	1
8.	a) Write a program in OpenGL to recursively subdivide a triangle to form 2D Sierpinski gasket. The number of recursive steps is to be specified by the user.	12	L3	5	1
	b) Describe specifically how each of these OpenGL primitives are used,				
	i) GL_LINES				
	ii) GL_LINE_LOOP				
	iii) GL_TRIANGLES				
	iv) GL_POLYGON	8	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
